

Parth Shah

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EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science (Specialization: AI & ML)

Chennai, India

Graduation: 2027 | CGPA: **7.55 / 10.0**

EXPERIENCE

Software Engineering Intern

Jun 2026 – Jul 2026

Manbal.ai

Bengaluru

- Acted as **product owner** for an internal CRM Engine, independently designing and shipping a full-stack lead management platform tracking **60+ institutions** and **122+ festival pipeline entries** with real-time status updates persisted to a cloud PostgreSQL database via Drizzle ORM
- Engineered a **role-based access control system** with three privilege tiers (SuperAdmin, Manager, Sales Executive), securing data routes at both the API middleware layer and frontend render level using **JWT authentication** and Zustand state management
- Rebuilt the frontend design system from scratch, delivering a **premium component library** — modal forms, drag-and-drop CSV import, role-differentiated UI, and a mobile-first responsive layout across **8+ workspace views** using React and TailwindCSS

Machine Learning Intern

Jan 2025 – Mar 2025

CODTECH IT Solutions

Hyderabad, India

- Engineered automated data preparation pipelines using Python and Pandas, reducing manual preprocessing effort by **30%**
- Developed and evaluated classification and regression models, improving prediction accuracy by **12%**

PROJECTS

PixelVault | Python, FastAPI, YOLOv8, AES-256-GCM, RSA-2048, React.js, PostgreSQL, Web Crypto API | [GitHub](#)

- Architected a Zero-Trust**, privacy-preserving image sharing system integrating **YOLOv8**-based ROI detection to selectively encrypt sensitive regions (**15%** of image), preserving full contextual visibility
- Engineered a hybrid cryptographic pipeline (AES-256-GCM + RSA-2048)** with browser-side decryption via Web Crypto API, ensuring complete **server-side blindness** to plaintext data

Distributed Order Processor | Python, FastAPI, RabbitMQ, PostgreSQL, Redis, Docker | [GitHub](#)

- Architected a scalable distributed order processing pipeline using **FastAPI** and **RabbitMQ** to decouple synchronous APIs and handle high-throughput traffic
- Engineered asynchronous background workers to process concurrent transactions with **fault tolerance** and zero data loss during simulated flash-sale testing
- Integrated Redis-based distributed caching** to reduce average data retrieval latency by **40%+**

Derma Vision AI | Python, TensorFlow, OpenCV, Supabase, Streamlit | [GitHub](#)

- Engineered an end-to-end computer vision pipeline** for real-time, high-fidelity skin type classification from user-uploaded images
- Architected a cloud-native backend** using Supabase (PostgreSQL) for secure user data and authentication

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, JavaScript, TypeScript

Frontend: React.js, TailwindCSS, HTML5, CSS3, Vite, Zustand

Backend & Databases: Node.js, Express.js, FastAPI, PostgreSQL, Redis, RabbitMQ, Drizzle ORM, REST APIs

AI & ML: TensorFlow, OpenCV, Pandas, YOLOv8, Scikit-learn

DevOps & Tools: Docker, Git, GitHub, Linux, Turborepo, JWT, Neon DB

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Distributed Systems, Information Security, RBAC

CERTIFICATIONS

Supervised Machine Learning: Regression and Classification | [DeepLearning.AI](#), [Coursera](#) | [Verify](#)